

Assignment 1

Section A: Data Fundamentals

① What is a database:

An organized collection of information that is stored electronically so that it can be easily accessed, managed, updated and protected

Example: A bank can use database to store customer details, account balances and transaction histories.

② What is a data warehouse.

A data warehouse is a central system that collects and stores data from different sources so that it can be analysed and used for reporting and decision making

Example: A bank can use a data warehouse to compare loans/credit scores of their clients, which one is more popular; if their clients have good/bad credit score.

③ What is data lake?

A data lake is a large, central storage system that can hold data in its original or raw form.

It can store structured, semi-structured and unstructured data

Example: raw customer data, names, id, address, phone numbers.

④ What is a Data Mart.

Is a smaller section or subset of a data warehouse that focuses on a particular department, business function or subject.

Example

The Finance department could use a finance data mart containing budgets, expenses etc.

⑥ Structured, Semi structured and unstructured data.

Structured data - Is information that follows a fixed structure, usually in rows and columns

Example: A table containing employee numbers, names, Salaries and departments.

Semi-structured - Does not follow a strict table structure but still contains some organisation or identifying information.

Example: A File containing customer information

Unstructured data - does not have a predefined structure.

It can include information such as text, photographs, audio and videos.

⑥ What is data Modeling

The process of creating a plan or representation showing what data an organisation needs, how the data is organised and how different pieces of data are related.

Two types of data models

Conceptual data Model - This provides a high level view of the important entities and relationships in a system, for example, a school might have students, teachers & courses

Logical data Model - This provides more detail about the data, including attributes and relationships, without necessarily specifying the exact technology that will store it.

7 What is data mining?

The process of examining large amounts of data to discover useful patterns, relationships, trends or other valuable information.

Example Fraud detection in banking.. A bank can analyse transaction patterns and identify unusual transaction patterns and identify.

8 What is database management system

Software that allows users and applications to create, store, organise, retrieve, update and manage information in databases.

4 Popular DBMS platforms.

① MySQL

Is a popular open-source relational database management system. It is commonly used for websites and web applications. Suitable use: An online store could use MySQL to manage customers, products and orders.

② PostgreSQL

An open-source relational database known for its reliability, flexibility and support for complex operations.

Suitable use: A financial application that requires reliable transactions could use PostgreSQL.

③ Oracle Database

Database is a powerful database platform commonly used by large organisations that require extensive database capabilities.

Suitable: A large bank / multinational company could use Oracle to manage large scale business data.

④ MongoDB

Is a non-relational document-oriented database. It stores information in flexible document structures rather than traditional relational tables.

Suitable use: A rapidly changing web/mobile application could use MongoDB when its data structure needs flexibility.

⑨ What is ETL?

ETL stands for Extract, Transform and Load. It is a process used to collect data from different sources, prepare it and move it into a destination such as a datawarehouse.

Extract - Data is collected from different sources, such as databases, spreadsheets, applications or APIs.

Transform - The data is cleaned and changed into a useful and consistent format. This could include removing duplicates, correcting errors or changing date formats.

Load - The prepared data is loaded into the target system, such as a data warehouse.

⑩ Common data storage formats.

CSV (Comma-separated values)

Stores information in a simple table-like format where values are separated by commas. It is easy to open in Excel and is useful for small or relatively simple datasets.

JSON (JavaScript Object Notation)

Stores information in a structured, key-value format. It is commonly used when transferring data between applications and through APIs.

Parquet

Is a column-oriented data format designed for efficient storage and retrieval. It is particularly useful for large analytical datasets because it supports efficient compression and encoding.

Avro

Data serialization format that uses schemas to describe the structure of the data. It is useful in data pipelines and streaming systems.

Section B: Artificial Intelligence & Machine Learning.

(i) What is Artificial Intelligence

Refers to technology that enables computers and machines to perform tasks that normally require aspects of human intelligence, such as learning, problem-solving, decision-making, understanding language and recognizing objects.

(ii) What is machine learning?

Is a branch of AI in which computer systems learn patterns from data and use those patterns to make predictions or decisions without being given explicit instructions for every possible situation.

(iii) What is deep learning and what are neural networks.

Is a specialised area of machine learning that uses neural networks with multiple layers to learn complex patterns from data.

Deep Neural Network - Is a computing model inspired loosely by the way biological brains process information.

⑭ What is natural language processing?

Is an area of AI that enables computers to work with human language, including written text and speech.

example: voice assistants.

⑮ What is a large language model

Is a type of deep learning model trained on very large amounts of data so that it can process and generate human language. example: ChatGPT.

⑯ What is generative AI

Is a type of artificial intelligence that can create new content based on patterns learned from training data.

It can generate text, images, audio, video and computer code.

It works through major stages - Training, tuning - Generation.

Example: Google Gemini - Can generate and work with text and other forms of content.

⑰ What is a training dataset?

Is a collection of examples that is used to teach a machine learning or AI model how to identify patterns and produce useful outputs.

⑱ What is computer vision

Computer vision is a field of AI that allows computers to process, analyse and interpret visual information such as photographs and videos.

⑲ What is an API (Application programming interface)

Is a set of rules and instructions that allows two different

Software applications or systems to communicate with each other.

② What is an API Key?

Is a unique code or identifier that an application uses when making requests to an API. It helps the API provider identify the application or user making the request and can be used to control access and monitor how the API is being used.

Example: Weather Service provides an API

The application → API request + API Key → Weather API →
→ Weather Information

Section C: APIs, Tools & Developer Concepts.

② What is a REST API

Is an application programming interface that follows the REST architectural style.

REST stands for Representational State Transfer. REST APIs allow different software applications to communicate with each other over the internet using HTTP.

② JSON - What is JSON?

JSON stands for JavaScript Object Notation. It is a lightweight format used to represent and exchange data between computer systems. JSON is popular because it is easy for both humans and computers to read and is supported by many programming languages.

Example JSON

```
{  
  "name": "Sedi"  
  "age": 22  
  "Occupation": "Data Analyst"  
}
```

(23) What is Prompt engineering?

Is the process of creating clear and effective instructions, called prompts, to guide an AI system towards producing a useful and appropriate response

Example

Poor prompt!

Tell me about Excel

Improved prompt

Explain five important Excel functions for a beginner data analyst. Give a simple explanation and one example for each function.

(24) What is a MAP function?

Is used in programming and data processing to apply the same function or operation to every item in a collection of data.

It takes each item, processes it and produces a new result.

(25) What is cloud computing?

Is a way of accessing computing resources such as storage, servers, applications and processing power over a network, usually the internet, instead of having to own and manage all the physical equipment yourself.

The three main cloud service models are IaaS, PaaS and SaaS.

(26) What is a version control system?

Is a system that records changes made to files over time.

It allows people to see previous versions of their work, compare changes and return to an earlier version when necessary.

Example : Git

(27) What is a Jupyter Notebook?

Is an interactive web-based environment where users can combine computer code, written explanations, equations, visualizations and results in one document,

(28) What is python?

Is a high-level programming language that is widely used for software development, automation, data analysis, Artificial Intelligence and machine learning.

Four python libraries commonly used in data work

- ① Pandas - used for working with tables, datasets and data cleaning
- ② Numpy - Used for numerical calculations and working with arrays
- ③ Matplotlib - Used to create graphs and data visualisations
- ④ Scikit-learn - Used for machine learning and predictive modeling

Section D: Data ethics, governance & careers

(29) What is data privacy?

- It's the principle that people should have control over their personal information, including how organisations collect, store, share and use it.

Example, ID's, financial information, & health information.

(30) What is data ethics?

Refers to the principles and responsible practices that guide how data and AI should be collected, analysed and used.

It involves considering whether the use of data is fair, responsible, transparent and respectful of people's rights

Three examples ^{OF} information unethical use of data/AI

- ① Using information / personal information without permission.
- ② Using biased AI for recruitment
- ③ Using AI to spread false information.

31) What is data governance?

Is the system of policies, rules, responsibilities and processes that an organisation uses to manage its data properly.

The purpose of data governance is to make sure that data is accurate, secure, available, consistent and used appropriately.

32) What is Bias in AI?

Occurs when an AI system produces unfair or distorted results because of bias in its training data, algorithms or the way the system was designed.

Example

Recruitment agency using AI to hire a certain demographic

33)

Role	Main responsibility	Important skills	Common tools
Data analyst	Examines data to find trends and provide business insights.	Excel, SQL, Statistics, Communication & Visualization	Excel, SQL, Power BI, Tableau, Python.
Data Scientist	Uses advanced statistics and machine learning to find patterns and make predictions	Python, statistics, machine learning & Maths	Python, R, SQL, Scikit learn.
Data Engineer	Builds and maintains systems and pipelines that collect, transform & store data	SQL, programming, databases, and Cloud technologies	SQL, Python, Spark, cloud platforms.
AI Engineer	Builds, integrates, deploys and maintains AI-powered applications and systems	Python, machine learning, APIs, cloud & software engineering	Python, TensorFlow, PyTorch, APIs, Git, cloud

34) What is business intelligence?

Refers to the processes and technologies businesses use to collect analyse and present data so that people can make better decisions.

Example BI can help organisations identify trends.

2 popular BI tools

1) Microsoft Power BI - Power BI is a business intelligence & data visualisation tool.

2) Tableau - data visualization & ~~Tableau~~ BI platform that help users explore data & create visual reports & dashboards.

Section E: Personal Reflection

35) Concept I found most interesting / surprising? WHY?

The concept I found interesting was Artificial Intelligence and AI bias. I found it surprising that an AI system can produce unfair results.

36) Which concept was difficult to understand? Resources used to understand?

The concept was REST API's and how different software applications communicate with each other.

I used documentation from websites such as AWS, MPM web docs and python documentation to understand the technical terms better.

37) How do I see data & AI affecting my future career?

I believe data and AI will have a major effect on my future career. because I am interested in finance and data analytics. I can use data analysis to examine financial information, identify trends, create reports and help businesses make better decisions.

Sources used for the Assignment.

- ① Amazon web Services (2026). What is RESTful API? Amazon web Services. AWS - what is RESTful API?
- ② MDN Web Docs. (2025). HTTP request methods. Mozilla Development Network. MDN - HTTP request methods.
- ③ IBM. (2026) What is prompt Engineering? IBM Think. IBM - prompt Engineering.
- ④ IBM (2023). What is AI Bias? IBM Think!
- ⑤ Python Software foundation. (2026) Built in functions.